# Advanced Basketball Intelligence Formulas (Oliver-style Frameworks)

Original summaries/notation capturing ideas widely used by analysts influenced by Dean Oliver. Calibrate coefficients to your league and data source.

## Notation

FG2M, FG3M, FGM, FGA, 3PA, FTM, FTA, ORB, DRB, TRB, AST, STL, BLK, TOV; Team_* for team totals; Opp_* for opponent totals; MP = minutes played.

## 1) Player Offensive Rating (ORtg) — template

$ORtg\_player = 100 \times PointsProduced / IndividualPossessions$

– Scale per 100 possessions used by the player.

## 1a) Points Produced (PP)

$PP = (2\times FG2M + 3\times FG3M) + FTM + Assist\_Points + Putback\_Points$

$Assist\_Points = k\_ast \times AST \times Team\_Points\_per\_Made\_FG$

– k_ast ~ 0.5–0.7; avoids double-counting.

$Team\_Points\_per\_Made\_FG = (2\times Team\_FG2M + 3\times Team\_FG3M) / Team\_FGM$

$Putback\_Points \approx k\_orb \times ORB\_putback\_FGM \times Avg\_Points\_per\_Putback$

## 1b) Individual Possessions (IPoss)

$ScPoss = FG\_ScPoss + FT\_ScPoss + ORB\_ScPoss$

$FG\_ScPoss = FGM$

$FT\_ScPoss \approx FT\_EndFraction \times FTA \times FT\%$

– FT_EndFraction ~ 0.44 for two-shot/triple-shot weighting; tune with play-by-play.

$ORB\_ScPoss = k\_orb \times ORB\_putback\_FGM$

$MissedFG\_NoTeamORB = (FGA - FGM) \times (1 - Team\_ORB\%)$

$MissedFT\_Ending \approx FT\_EndFraction \times FTA \times (1 - FT\%) \times \phi$

– $\phi$ fraction of FT situations that end the possession if missed.

$IPoss = ScPoss + MissedFG\_NoTeamORB + MissedFT\_Ending + TOV$

## 2) Assist Credit Models

$Assist\_Points\ (simple) = k\_ast \times AST \times Team\_Points\_per\_Made\_FG$

$Assist\_Points\ (contextual) = \Sigma\_assisted\_shots\ [\ k\_ast(z) \times points(z)\ ]$

– z indexes zone/type; k_ast varies with shot value.

## 3) Offensive Rebound Credits

$Team\_ORB\% = ORB / (ORB + Opp\_DRB)$

$Putback\_Rate = Putback\_FGA / ORB\ ;\ Putback\_PPS = Points\_putbacks / Putback\_FGA$

$Putback\_Points \approx k\_orb \times ORB \times Putback\_Rate \times Putback\_PPS$

## 4) Player Defensive Rating (DRtg) via Stops

$Stops = STL + k\_blk\times BLK\_recovered + k\_drb\times DRB + Forced\_Miss\_Credit + Team\_Stop\_Share$

$Stop\% = Stops / Defensive\_Poss\_on\_Floor$

$Team\_Def\_PPP\_on\_Floor = Opp\_Points\_on\_Floor / Defensive\_Poss\_on\_Floor$

$DRtg\_player = 100 \times Team\_Def\_PPP\_on\_Floor \times (1 - Stop\_Adjustment)$

– Stop_Adjustment derived from Stop% vs team baseline.

$Team\_Stop\_Share = k\_team \times (Team\_Stops - \Sigma\ individual\ components)$

## 5) Possessions & Ratings (team/lineup)

$Poss\_team \approx 0.5\times[(FGA + 0.44\times FTA - ORB + TOV) + (Opp\_FGA + 0.44\times Opp\_FTA - Opp\_ORB + Opp\_TOV)]$

$OffRtg = 100 \times Points / Poss\_team$ ; $DefRtg = 100 \times Opp\_Points / Opp\_Poss$ ; $NetRtg = OffRtg - DefRtg$

$Lineup\_NetRtg = 100 \times (Pts\_for - Pts\_against) / Poss\_lineup$

## 6) Four Factors Decomposition (delta form)

$eFG = (FGM + 0.5 \times FG3M) / FGA$ ; $FTr = FTA / FGA$ ; $ORB\% = ORB / (ORB + Opp\_DRB)$ ; $TOV\% = TOV / Poss$

$\Delta OffRtg \approx 100 \times [\ w1\Delta eFG + w2\Delta FTr + w3\Delta ORB\% - w4\Delta TOV\%\ ]$

– Choose weights w1..w4 to reflect leverage in your era.

## 7) Opponent & Luck Adjustments

$Adj\_OffRtg(team) = OffRtg - E\_vs\_ops + League\_Avg$

– E_vs_ops is possession-weighted expectation vs opponent defensive strengths.

$LuckAdj\_3P\% = \lambda \times Expected\_3P\% + (1-\lambda) \times Observed\_3P\%$

– Expected from shot quality/contest; $\lambda$ controls regression-to-mean.

*LuckAdj_Def_3P%: replace opponent makes with xMake_prob from contest models to remove variance.*

## 8) Pythagorean Expectation Tuning

$Win\%\_x = PF^x / (PF^x + PA^x)$ ; $ExpectedWins = Win\%\_x \times Games$

$x^* = \mathrm{argmin}_x\ \Sigma\ (Win\%\_x - ActualWin\%)^2$

– Fit exponent x by minimizing squared error for a season/era.

## 9) Usage–Efficiency & Skill Curves

$USG\% \approx 100 \times (FGA + 0.44 \times FTA + TOV) \times (Team\_MP / (5 \times Player\_MP \times Team\_Poss))$

$Eff(USG) = a - b \times USG$ or $Eff(USG) = A - B \times USG^\rho\ (\rho > 1)$

– Concave response models diminishing returns with higher usage.

Team production $f(u1,...,un)$ concave: $\partial^2 f/\partial u\_i^2 < 0$ ; optimize $\Sigma u\_i = Team\_Usage$

– Solve with Lagrange/KKT or numerical search.

## 10) On/Off, WOWY, and RAPM-style Estimation

$OnOff = NetRtg\_on - NetRtg\_off$ ; $WOWY(A,B) = NetRtg\_\{A,B\} - NetRtg\_\{A,\neg B\}$

RAPM: β■ $= \mathrm{argmin}\_\beta\ ||y - X\beta||^2 + \lambda||\beta||^2$

– y: possession outcomes; X: player-on indicators; $\lambda$: ridge penalty to stabilize.

## 11) Calibration & QA Notes

- Tune k_ast, k_orb, k_team so player sums approach team totals; report residuals.
- Refine FT ending weights using play-by-play (and-1s, technicals, lane ORBs).
- Keep a data dictionary: define what counts as a possession for each stat.

— End —